

MATHEMATICS 5310

Hints and solutions to selected exercises from 2.5

1. (a) Yes. $\text{Ker} = \mathbf{Z}_n$. (b) No. It is a homomorphism only if G is abelian. In this case $\text{Ker} = \{e\}$. (c) Yes. $\text{Ker} = \{e\}$. (d) Yes. $\text{Ker} = \mathbf{R}_{>0}$. (e) Yes. 3. (a) L_a is injective: If $L_a(x) = L_a(y)$, then $xa^{-1} = ya^{-1}$ and by right cancellation, $x = y$. L_a is surjective: If $y \in G$, then $L_a(ya) = yaa^{-1} = ye = y$.

(b) $L_a L_b(x) = L_a(xb^{-1}) = xb^{-1}a^{-1} = x(ab)^{-1} = L_{ab}x$.

(c) If $\psi(a) = id$, then $xa^{-1} = x$ for any $x \in G$. By left cancellation $a^{-1} = e$ and hence $a = e$.

4. $T_a L_b(x) = T_a(xb^{-1}) = a(xb^{-1}) = (ax)b^{-1} = L_b(ax) = L_b T_a(x)$.

5. If $T_a V = V T_a$, then $T_a V(e) = V T_a(e)$. Let $b = V(e)$, then $T_a(b) = V(a)$ and so $ab = V(a)$ i.e. $V(a) = L_{b^{-1}}(a)$.

6. Closed under $*$: If $g', h' \in \varphi(G)$, then $g' = \varphi(g)$ and $h' = \varphi(h)$ for some $g, h \in G$. Thus $g'h' = \varphi(g)\varphi(h) = \varphi(gh) \in \varphi(G)$.

Closed under inverses: If $g' \in \varphi(G)$, then $g' = \varphi(g)$ for some $g \in G$. Thus $(g')^{-1} = \varphi(g)^{-1} = \varphi(g^{-1}) \in \varphi(G)$.

7. Clearly if φ is injective, then $\text{Ker}(\varphi) = \{e\}$.

Conversely, assume that $\text{Ker}(\varphi) = \{e\}$. If $\varphi(x) = \varphi(y)$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y^{-1}) = \varphi(x)\varphi(y)^{-1} = e$ and so $xy^{-1} \in \text{Ker}(\varphi) = \{e\}$. Hence $xy^{-1} = e$ so that $x = y$.

8. Define $f : (\mathbf{R}, +) \rightarrow (\mathbf{R}_{>0}, \cdot)$ by $f(x) = e^x$. We know that f is strictly increasing, continuous and $\lim_{x \rightarrow -\infty} = 0$, $\lim_{x \rightarrow \infty} = \infty$. Thus, f is bijective. Since $e^{x+y} = e^x e^y$ it is an isomorphism.

10. Clearly $gHg^{-1} = H$. Since $fgf^{-1} = g^3$ it is easy to see that $fHf^{-1} = H$. Since any element in G is of the form $f^i g^k$, it is easy to see that H is normal in G .

11. Similar to 10.

12. If $z \in Z(G)$ and $g \in G$, then $zg = gz$. Thus $gzg^{-1} = z$ and hence $gZ(G)g^{-1} \subset Z(G)$.

13. Let n be the least common multiple of the orders of elements of G . (One can see that in fact n is the biggest order of an element of G .)

If $(n, m) = 1$, then $an + bm = 1$ for some $a, b \in \mathbf{Z}$. Since φ is a homomorphism and G is finite, it is an isomorphism iff it is injective iff it is surjective. Suppose that $g^m = \varphi(g) = e$, then $g = g^1 = g^{an+bm} = (g^n)^a (g^m)^b = e^a e^b = e$ i.e. φ is injective.

Suppose that φ is injective. Let $(n, m) = c$. We will show that $c = 1$. If $c \neq 1$, then $d = n/c < n$ and there is a $g \in G$ such that $g^d \neq e$. Since φ is an isomorphism, $g = h^m$ and so $e \neq g^d = h^{dm} = h^{nm/c} = (h^{m/c})^n = e$ which is impossible.

14. $g'h' = \varphi(g)\varphi(h) = \varphi(gh) = \varphi(hg) = \varphi(h)\varphi(g) = h'g'$.

15. Let $N' = \varphi(N)$ and $g' \in G'$. We must show $g'N'(g')^{-1} \subset N'$. Let $n' \in N'$, then $n' = \varphi(n)$ and $g' = \varphi(g)$ for some $n \in N$ and $g \in G$. We have $g'n'(g')^{-1} = \varphi(g)\varphi(n)\varphi(g^{-1}) = \varphi(gng^{-1}) = \varphi(n_0)$ where $n_0 \in N$ and so $g'n'(g')^{-1} \in N'$.

16. Closed under $*$: If $m \in M$ and $n \in N$, then $nmn^{-1} = m_0$ for some $m_0 \in M$ (by normality of M) thus $nm = m_0n$. If $m, m' \in M$ and $n, n' \in N$, then $mnmm'n' = mm'_0nn' \in MN$.

Closed under inverses: If $m \in M$ and $n \in N$, then $m^{-1} \in M$ and $n^{-1} \in N$. Now $(mn)^{-1} = n^{-1}m^{-1} = m_0n^{-1} \in MN$.

MN is normal in G : $gmng^{-1} = gmng^{-1}gnng^{-1} = m_0n_0 \in MN$.

17. Similar to 16.

18. We must show $g^{-1}Ng \subset N$ for any $g \in G$. let $n \in N$, then $g^{-1}ng \in \cap_{a \in G} g^{-1}a^{-1}Hag = \cap_{a \in G} (ag)^{-1}H(ag) = \cap_{b \in G} b^{-1}Hb = N$ (we have renamed ag by b and observed that $G = \{ag | g \in G\}$).

19. a) Closure under $*$: If $n, m \in N(H)$, then $n^{-1}Hn = H$ and $m^{-1}Hm = H$. But then $(nm)^{-1}H(nm) = m^{-1}n^{-1}Hnm = m^{-1}Hm = H$. Closure under inverses: If $n \in N(H)$, then $n^{-1}Hn = H$ so $nHn^{-1} = nn^{-1}Hnn^{-1} = eHe = H$.

b) By definition of $N(H)$.

c) If H is normal in K , then for any $k \in K$, we have $k^{-1}Hk = H$ so that $k \in N(H)$ and hence $K \subset N(H)$.

20. $n^{-1}m^{-1}nm = m'm \in M$ as M is normal in G and $n^{-1}m^{-1}nm = n^{-1}n' \in N$ as N is normal in G . Hence $n^{-1}m^{-1}nm \in M \cap N$ so $n^{-1}m^{-1}nm = e$ so $mn = mne = mn(n^{-1}m^{-1}nm) = (mnn^{-1}m^{-1})nm = enm = nm$.

21. Pick $t \neq s$ and let $f \in A(S)$ be the map that switches s and t . It is easy to see that $f^{-1}H(s)f = H(t) \neq H(s)$.

22. $f = (1, 2)$. The right cosets are $He = H(1, 2) = \{e, (1, 2)\}$; $H(1, 3) = H(1, 3, 2) = \{(1, 3), (1, 3, 2)\}$ and $H(2, 3) = H(1, 2, 3) = \{(2, 3), (1, 2, 3)\}$. The left cosets are $eH = (1, 2)H = \{e, (1, 2)\}$; $(1, 3)H = (1, 2, 3)H = \{(1, 3), (1, 2, 3)\}$ and $(2, 3)H = (1, 3, 2)H = \{(2, 3), (1, 3, 2)\}$.

23. Since (a) is a subgroup, it is a normal subgroup by assumption and hence $bab^{-1} \in (a)$ i.e. $bab^{-1} = a^j$ so that $ba = a^jb$.

29. a) Let $w \in W$ and let ϕ_w be the automorphism of W defined by conjugation: $\phi_w(x) = wxw^{-1}$. Since M is characteristic, $\phi_w(M) \subset M$ for all $w \in W$. But then $wMw^{-1} \subset M$ for all $w \in W$ so that M is normal in W .

b) Let ϕ be an automorphism so that $\phi(M) \subset M$ and $\phi(N) \subset N$. Let $n \in N$ and $m \in M$, then $\phi(mn) = \phi(m)\phi(n) = m'n' \in MN$. So $\phi(MN) \subset MN$. We must also check that MN is a subgroup. This follows from part a and ex. 16.

43. We may assume that G is not cyclic and hence that the order of all non-identity elements is 3. Therefore there are 4 subgroups of order 3. Conjugation defines a homomorphism $G \rightarrow S_4$. Since 9 does not divide $4!$, there is a nontrivial element g^i in the kernel, i.e. an element g such that $g^{-1}Hg = H$ for any subgroup H of G . Let $x \neq g^i$ for any i , then $gxg^{-1} \in \langle x \rangle$ and one can see that either $gxg^{-1} = x$ or $gxg^{-1} = x^2$. In the first case, the group is abelian (in fact $\mathbf{Z}_3 \times \mathbf{Z}_3$). In the second case, $z = exe = g^3zg^{-3} = g^2z^2g^{-2} = gz^4g^{-1} = gzg^{-1} = z^2$ is impossible.

44. As in the previous exercise, every non identity element has order p and there must be $(p+1)$ subgroups of order p . Conjugation then defines a homomorphism $G \rightarrow S_{p+1}$. Since p^2 does not divide $(p+1)!$, there is an element g in the kernel. Let z be an element of $G - \langle g \rangle$, then $gzg^{-1} = z^i$ for some $1 \leq i \leq p-1$. We have $z = g^p z g^{-p} = z^{i^p}$. Since the order of z is p and since i^p is congruent to i modulo p (FLT), we obtain $z = z^i$ and so $i = 1$. But then $gzg^{-1} = z$ and so $G \cong \mathbf{Z}_p \times \mathbf{Z}_p$.